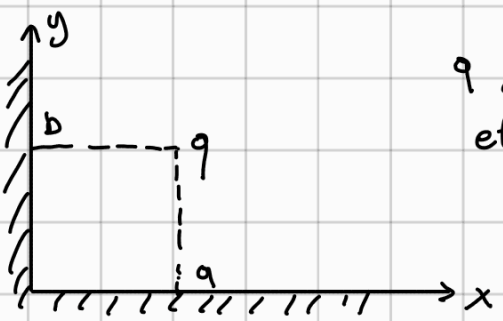
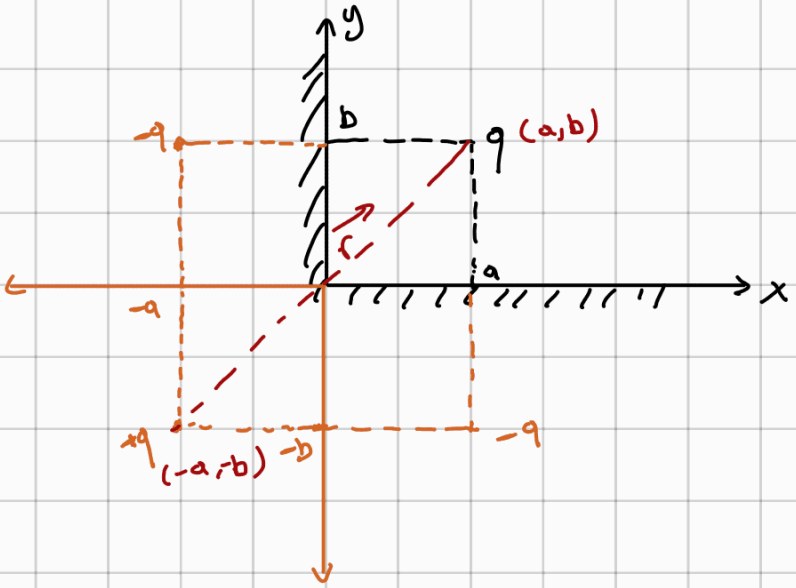
görüntü teoremi

SGS Pr 6



q yüküne etki eden kuvvet



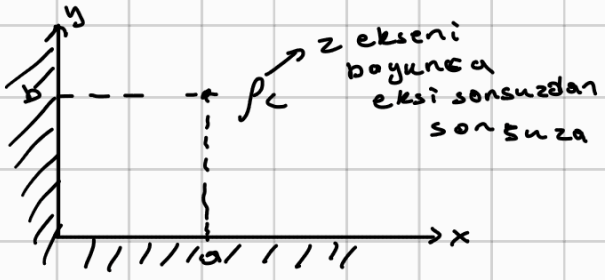
$$\vec{F} = -\frac{q^2}{4\pi\varepsilon 4a^2}\vec{e}_x - \frac{q^2}{4\pi\varepsilon 4b^2}\vec{e}_y + \frac{q^2}{4\pi\varepsilon 2\sqrt{a^2+b^2}}\left(\frac{a\vec{e}_x + b\vec{e}_y}{\sqrt{a^2+b^2}}\right)$$

$$|\vec{r}| = 2\sqrt{a^2+b^2}$$

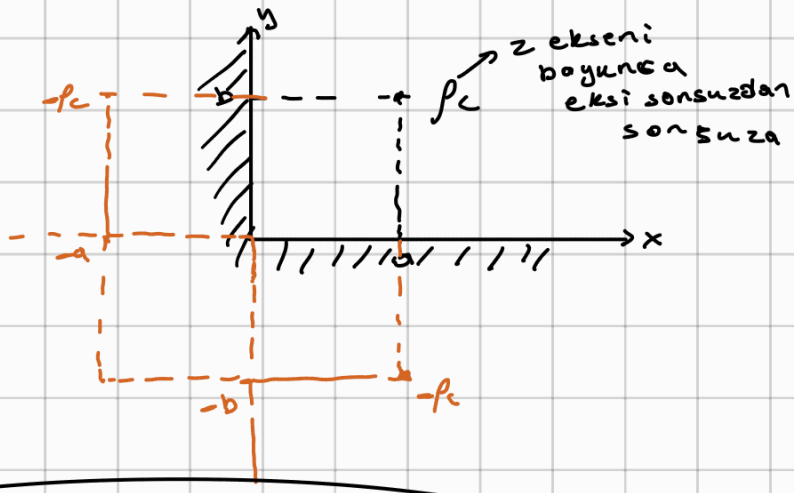
$$\vec{e}_r = \frac{\vec{r}}{|\vec{r}|} = \frac{2a\vec{e}_x + 2b\vec{e}_y}{2\sqrt{a^2+b^2}} = \frac{a\vec{e}_x + b\vec{e}_y}{\sqrt{a^2+b^2}}$$

$$\vec{F} = \frac{q^2}{16\pi\varepsilon}\left[\vec{e}_x\left(-\frac{1}{a^2} + \frac{a}{(a^2+b^2)^{3/2}}\right) + \vec{e}_y\left(-\frac{1}{b^2} + \frac{b}{(a^2+b^2)^{3/2}}\right)\right]$$

SGS Pr 7



ρ_c'nin birim uzunluğuna etki eden kuvvet nedir?



Sonsuz uzun çizgisel yük dağılımı için

$$\vec{E} = \frac{\rho_e}{2\pi\varepsilon\rho}\vec{e}_\rho$$

$$\vec{E} = -\frac{\rho_c}{2\pi\varepsilon 2a}\vec{e}_x - \frac{\rho_c}{2\pi\varepsilon 2b}\vec{e}_y + \frac{\rho_c}{2\pi\varepsilon 2\sqrt{a^2+b^2}}\left(\frac{a\vec{e}_x + b\vec{e}_y}{\sqrt{a^2+b^2}}\right)$$

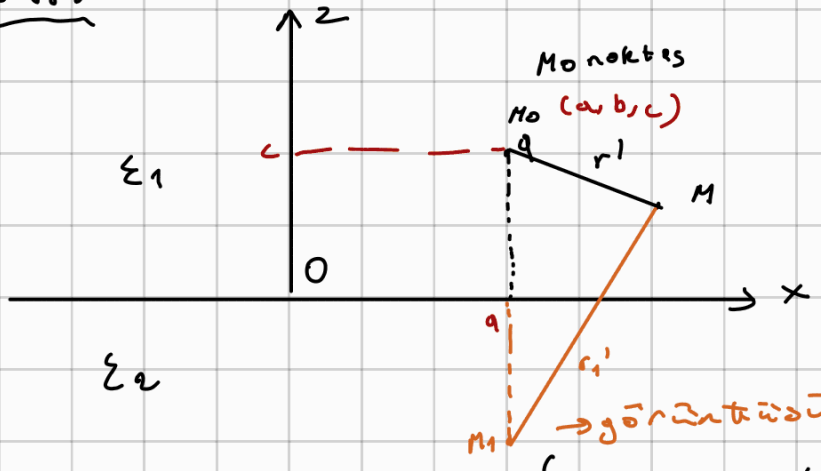
$$\vec{E} = \frac{\rho_c}{4\pi\varepsilon}\left(\vec{e}_x\left(-\frac{1}{a} + \frac{a}{a^2+b^2}\right) + \vec{e}_y\left(-\frac{1}{b} + \frac{b}{a^2+b^2}\right)\right)$$

$$= \frac{-\rho_c}{4\pi\varepsilon(a^2+b^2)}\left[+\vec{e}_x\frac{b^2}{a} + \vec{e}_y\frac{a^2}{b}\right]$$

$$\vec{F} = \int_{z=0}^{z=1}\rho_c\vec{E}\,dz = \rho_c\vec{E}$$

$$= \frac{-\rho_c^2}{4\pi\varepsilon(a^2+b^2)}\left[+\vec{e}_x\frac{b^2}{a} + \vec{e}_y\frac{a^2}{b}\right]$$

S66 P.8



q yükünün oluşturduğu pot. konksiyonu

$$V(M) = \begin{cases} \dfrac{q}{4\pi\varepsilon_1 r'} + \dfrac{q(\varepsilon_1-\varepsilon_2)}{4\pi\varepsilon_1(\varepsilon_1+\varepsilon_2)r_1'}, & z>0 \\ \dfrac{2q\varepsilon_2}{4\pi\varepsilon_2(\varepsilon_1+\varepsilon_2)r'}, & z<0 \end{cases}$$

olduğunu gösteriniz

$z=0$ düzleminde biriken yüke benzer.

Denk problem çözülmeden yazılır

$z>0 \quad \Delta V = \frac{-\rho_v}{\varepsilon_1} = -\frac{q}{\varepsilon_1}\,\delta(x-a)\,\delta(y-b)\,\delta(z-c)$

$z<0 \quad \Delta V = 0$

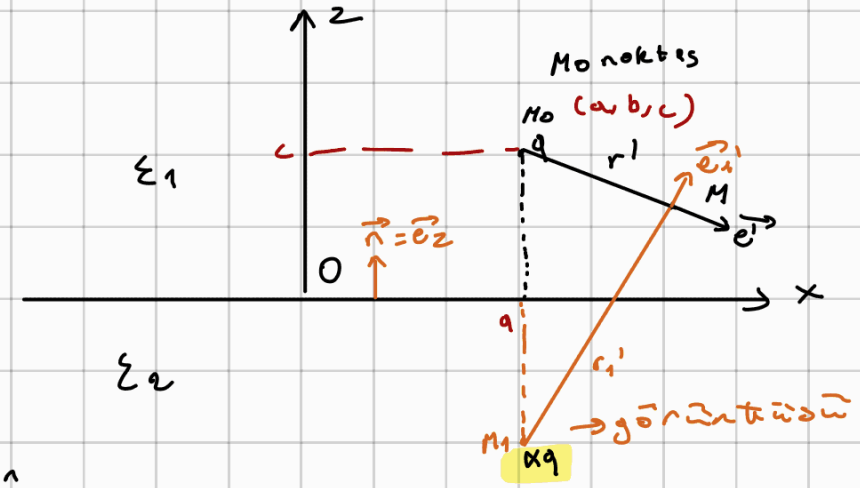
$z=0$ sınır problemleri;

$V_1 = V_2 \qquad \vec{n}\cdot\vec{D}_1 = \vec{n}\cdot\vec{D}_2$

$\vec{e}_z\cdot\vec{D}_1 = \vec{e}_z\,\vec{D}_2$

Şimdi denk problem den bakalım

$z>0 \quad \varepsilon_2 = \varepsilon_1$ seçelim



M'nin
$z>0$ iken

M noktasında kaynaklanan potansiyel

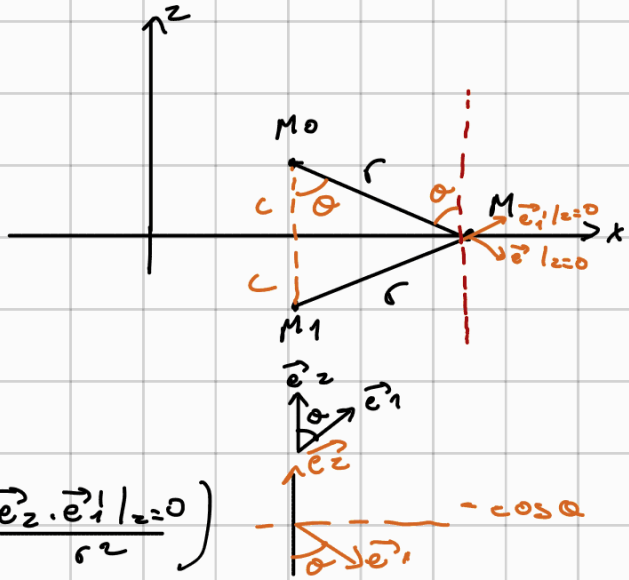
$$V_1 = \frac{q}{4\pi\varepsilon_1 r'} + \frac{\alpha q}{4\pi\varepsilon_1 r_1'}$$

$$\vec{E}_1 = -\text{grad}\,V_1 = \frac{q}{4\pi\varepsilon_1}\left(\frac{1}{r'^2}\vec{e}' + \frac{\alpha}{r_1'^2}\vec{e}_1'\right)$$

$$V_1|_{z=0} = \frac{q}{4\pi\varepsilon_1}\left(\frac{1}{r} + \frac{\alpha}{r}\right)$$

$$= \frac{q}{4\pi\varepsilon_1 r}\left(\frac{1}{r} + \frac{\alpha}{r}\right)$$

$$= \frac{q}{4\pi\varepsilon_1 r}(1+\alpha)$$

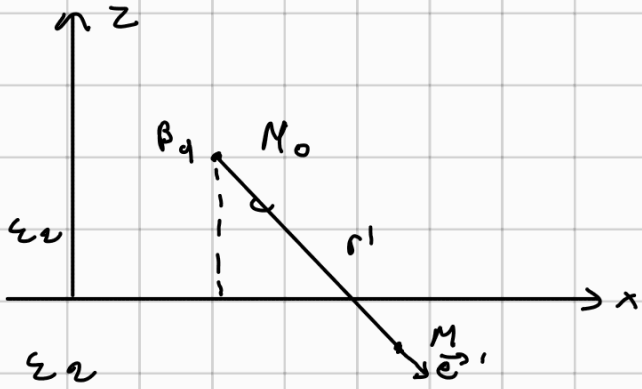


$\vec{e}_z\cdot\vec{D}_1\big|_{z=0}$ eşittir $= \frac{q}{4\pi}\left(\frac{\vec{e}_z\cdot\vec{e}'|_{z=0}}{r^2} + \frac{\alpha\,\vec{e}_z\cdot\vec{e}_1'|_{z=0}}{r^2}\right)$

$$= \frac{q}{4\pi r^2}\left(-\cos\theta + \alpha\cos\theta\right)$$

$$\boxed{= \frac{q\,c}{4\pi r^3}\,(\alpha - 1)}$$

$z < 0$ bölgesinde $\varepsilon_1 = \varepsilon_2 =$ olduğuna varsayarak M_0'a bir βq yükü yerleştirerek denk problemi tanımlayalım

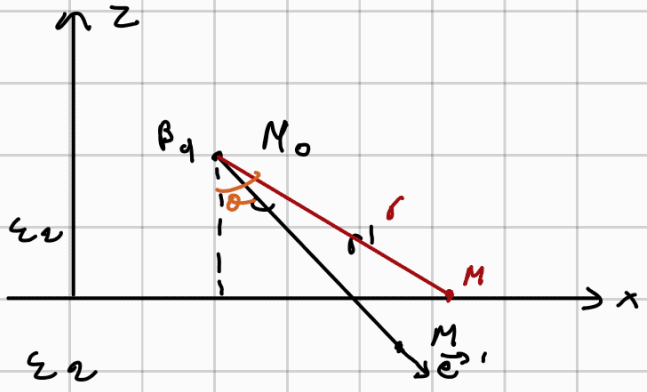


$$V_2 = \frac{\beta q}{4\pi\varepsilon_2 r'}$$

$$\vec{E} = -\text{grad}\,V$$

$$= -\frac{\partial V}{\partial r}\vec{e}'$$

$$= \frac{\beta q}{4\pi\varepsilon_2 r'^2}\vec{e}'$$



$z=0$ $\quad V_2|_{z=0} = \frac{\beta q}{4\pi\varepsilon_2 r}$

$$\vec{e}_z \cdot \vec{D}_2|_{z=0} = \frac{\beta q}{4\pi r^2}\vec{e}_z\vec{e}'|_{z=0}$$

$$= -\frac{\beta q}{4\pi r^2}\cos\theta$$

$$= -\frac{\beta q\,c}{4\pi r^3}$$

$$V_1|_{z=0} = V_2|_{z=0}$$

$$\frac{q}{4\pi\varepsilon_1 r}(1+\alpha) = \frac{\beta q}{4\pi\varepsilon_2 r}$$

$$\Rightarrow \boxed{\varepsilon_2(1+\alpha) = \varepsilon_1\beta}$$

$$\vec{e}_z \cdot \vec{D}_1|_{z=0} = \vec{e}_z \cdot \vec{D}_2|_{z=0}$$

$$\Rightarrow \frac{q\,c}{4\pi r^3}(\alpha - 1) = \frac{-\beta q\,c}{4\pi r^3}$$

$(\alpha - 1) = -\beta \Rightarrow \boxed{1 - \alpha = \beta}$

$\Rightarrow \alpha = \dfrac{\varepsilon_1 - \varepsilon_2}{\varepsilon_1 + \varepsilon_2}$

$\beta = \dfrac{2\varepsilon_2}{\varepsilon_1 + \varepsilon_2}$

$V_1 = \dfrac{q}{4\pi\varepsilon_1 r_1} + \dfrac{\alpha q}{4\pi\varepsilon_1 r_2}$

$$V(M) = \begin{cases} V_1, & z > 0 \\ V_2, & z < 0 \end{cases} = \begin{cases} \dfrac{q}{4\pi\varepsilon_1 r_1} + \dfrac{q(\varepsilon_1 - \varepsilon_2)}{4\pi\varepsilon_1(\varepsilon_1+\varepsilon_2) r_2}, & z > 0 \\ \dfrac{2q\varepsilon_2}{4\pi\varepsilon_2(\varepsilon_1+\varepsilon_2) r_1}, & z < 0 \end{cases}$$

$z = 0$ düzleminde biriken yükü bulalım

$\Delta V = \dfrac{-\rho V(x,y,z)}{\varepsilon} \qquad V|_{z=0} = V_0$

$V = f(x,y,z) - f(x,y,-z) + V_0$

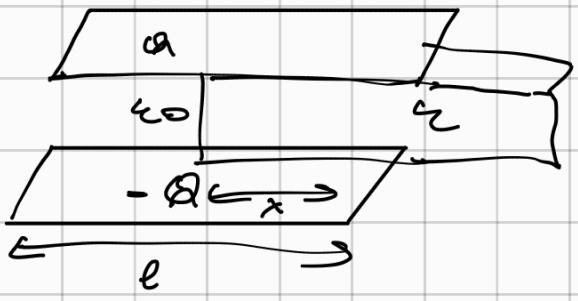
$\Delta V = \Delta f(x,y,z) - \Delta f(x,y,-z) + \Delta V_0$ (0, 0)

sadece $z > 0$ potansiyelini etkisini

$\rho(x,y,z) = -\varepsilon_1 \Delta V|_{z>0}$

Enerji yoğunlukları

Sayfa 7 Ü Poz



Sal, 1.3